Diabetes Prediction Project

Analysis and Design Phase | DEPI

This stage analyzes the team-supplied diabetes dataset and turns the agreed project scope into a practical design. The project will classify the recorded diabetes label (0/1) using the available demographic, lifestyle, and health fields. Dataset source, sample size, collection period, and label definition must be confirmed from the team's data documentation before final results are interpreted.

1. Project Analysis

The analysis establishes what the project can answer with the supplied data, who is involved, and whether the work is feasible within the team's academic schedule and available tools.

1.1 Problem Statement and Needs Assessment

Problem statement: The team has a dataset containing health-related fields and a binary diabetes label, but needs a documented and reproducible way to examine the data and compare classification models. Without checking data quality, label construction, and class balance, a model's reported performance may be misleading.

Felt needs: The team needs to understand the dataset, identify data-quality issues, compare suitable models, explain the results, and deliver a notebook, report, and presentation on schedule.

Normative needs: The project must follow the course proposal requirements, use a clear train/test evaluation method, report appropriate classification metrics, document assumptions and limitations, and avoid presenting an educational model as a medical diagnostic tool.

1.2 Stakeholder Analysis

Stakeholder	Influence / Interest	Main expectations
DEPI instructor / evaluator	High / High	A complete analysis and design, clear evidence for decisions, and deliverables aligned with the course requirements.
Project team	High / High	A feasible workload, clear responsibilities, reproducible analysis, and understandable results.
People represented in the dataset	Low direct influence / High interest	Responsible handling of the data and no unsupported medical claims. The dataset documentation and any privacy conditions must be followed.
Academic audience / presentation viewers	Medium / Medium	A clear explanation of the question, model comparison, evaluation, and limitations.

1.3 Feasibility Study

Technical feasibility: The proposed workflow can be implemented with Python, pandas, and scikit-learn in Jupyter Notebook or Google Colab. No specialized hardware is required. Feasibility depends on the dataset being readable and its fields and target being sufficiently documented.

Operational feasibility: The five-person team has a four-week project schedule and previously assigned roles for data preparation, model development, analysis, documentation, and presentation. The planned methods are suitable for a student project, provided work is coordinated around the 6 November analysis-and-design milestone.

Economic feasibility: No project budget or paid service is specified. The design relies on the dataset already supplied by the team and commonly available open-source tools; any required cloud storage or paid service should be checked before use.

2. Project Design

The design defines measurable project boundaries, the planned data and model workflow, the schedule, and safeguards for likely risks.

2.1 Project Objectives and Scope Constraints

SMART objective: By 4 December 2026, prepare a reproducible notebook that cleans and analyzes the team-supplied dataset, compares Logistic Regression, Decision Tree, and Random Forest for classifying the recorded diabetes label, and reports the results with a confusion matrix, accuracy, precision, recall, F1-score, and ROC-AUC where appropriate.

In scope: Inspecting the supplied fields; documenting dataset limitations; checking missing values, duplicates, ranges, and target balance; preprocessing; comparing the three selected classifiers; interpreting evaluation results; preparing the notebook, report, and presentation.

Out of scope: Collecting new patient data; diagnosing or treating individuals; claiming that the model predicts future disease onset without longitudinal outcome data; deploying a clinical system; and claiming clinical validation or performance beyond the tested dataset.

2.2 Technical and System Architecture

Layer	Planned function
Data input	Load the team's original dataset; preserve a copy; document source, sample size, collection period, fields, and target definition when available.
Validation and preparation	Check schema, duplicates, missing values, impossible or unexpected values, and class balance. Encode categorical fields; split data before fitting preprocessing; fit transformations on training data only.
Model training	Train Logistic Regression as a baseline, Decision Tree as an interpretable comparison, and Random Forest as a model able to capture nonlinear patterns.
Evaluation	Use a stratified holdout test set; report confusion matrix, accuracy, precision, recall/sensitivity, F1-score, and ROC-AUC when applicable. Add cross-validation if time and data size allow.
Outputs	Save a reproducible notebook, model comparison table, charts, written interpretation, limitations, and final presentation.

Important design check: Confirm how the diabetes label was created. If HbA1c or blood glucose values directly define that label, using them as predictors may cause target leakage or near-direct prediction. The team should document this and interpret results accordingly.

2.3 Timeline and Risk Matrix

Phase / date	Planned work and output
By 6 Nov 2026 - Analysis and design	Confirm data fields and target definition; assess data quality and class balance; document stakeholders, feasibility, scope, workflow, candidate models, and risks.
By 30 Nov 2026 - Implementation	Preprocess the data; train and compare the planned classifiers; prepare evaluation tables, charts, and an initial results summary.
By 4 Dec 2026 - Testing and final delivery	Rerun the notebook, verify reproducibility, explain limitations, complete the report, and prepare the presentation.

Risk event	Likelihood	Mitigation
Unclear target definition or leakage from HbA1c / glucose	Medium	Review the dataset documentation; verify how the label was created; compare results with and without a potentially label-defining field if needed; state the limitation.
Imbalanced diabetes classes	Medium	Check class proportions before training; use a stratified split; report recall and F1-score; apply any resampling only within training data.
Missing, duplicated, or inconsistent records	Medium	Run documented quality checks; record cleaning decisions; avoid silently dropping large parts of the dataset.
Overfitting or unstable performance	Medium	Keep a separate test set, use cross-validation if feasible, and compare multiple metrics rather than accuracy alone.
Schedule or coordination delays	Low to Medium	Assign an owner to each deliverable, share a common notebook,

		and review progress before each milestone.
Privacy or unsupported clinical interpretation	Low / to be verified	Follow the dataset’s access conditions; avoid including identifiable records; label the result as an educational classification prototype.